

A 28-GHz Flip-Chip Packaged Chireix Transmitter With On-Antenna Outphasing Active Load Modulation

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Abstract—This paper presents a packaged 28-GHz Chireix outphasing transmitter (TX) for power amplifier (PA) efficiency enhancement at both peak and back-off output power (PBO). The main objective is to investigate the possibility of employing a multi-feed antenna as a multi-port passive network to perform outphasing active load modulation. As a proof of concept, an outphasing TX architecture is proposed by leveraging a dual-feed loop antenna (DLA) to achieve high efficiency transmission of wideband complex modulated signals. Theoretical analysis and simulation results prove that the proposed DLA is electrically equivalent to an ideal differential series non-isolating power combiner, enabling superior on-antenna outphasing operation. Multi-functional signal processing is therefore achieved in a single loop-antenna footprint, including high-efficiency on-antenna power combining, outphasing active load modulation, and millimeter-wave (mm-Wave) signal radiation. Moreover, unlike conventional spatial vectorial combining techniques (e.g., spatial IQ or spatial outphasing), the antenna-level signal processing preserves the wide antenna unit field of view (FoV) and is particularly suitable for array or multiple-input-multiple-output (MIMO) applications. The proposed Chireix TX contains two phase-shifted sub-TXs simultaneously driving a DLA on package. The measurements demonstrate a 53%/43% PA drain/power-added efficiency (DE/PAE) at 17.1-dBm saturated output power (P_{sat}) and 36%/23% DE/PAE at 6-dB PBO. The high-efficiency Chireix TX supports up to 15 Gb/s 64-QAM with 10.4-dBm average output power and 34% average PA DE (19.2% average PAE) while maintaining EVM/ACLR better than -27.5 dBc/ -28 dBc in a wireless radiation testing environment.

Index Terms—5G, Chireix, CMOS, millimeter wave (mm-Wave), outphasing, power amplifier (PA), transmitter (TX), wireless communication.

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I. INTRODUCTION

THE exponential growth in data rates for modern wireless communication demands greater bandwidth for the upcoming 5G wireless systems. Millimeter-wave (mm-Wave) offers broader available spectrums and proportionate increase of channel capacity [1], [2]. Therefore, it is leveraged in 5G systems to address the increasing demands for fast data speed and low network latency. It also enables large-scale beamforming to compensate the high path loss (PL) at mm-Wave [3], [4]. Viable mm-Wave transmitter (TX) frontend solutions should handle multi-gigabit per second spectrum-efficient modulated signals, such as high-order quadrature amplitude modulations (QAMs) [5], [6]. Their corresponding large peak-to-average power ratios (PAPRs) place a demanding requirement on both peak and power back-off (PBO) efficiency to guarantee a high average efficiency in modulation. Moreover, high system power efficiency is of paramount importance to reduce cooling needs in massive multiple-input-multiple-output (MIMO) and large-scale phased-array systems, which will significantly reduce maintenance cost and ease system deployment.

Mm-Wave Doherty TX/PA is a widely used technique to enhance PA PBO efficiency [7]–[12]. However, many existing Doherty TX/PA designs exhibit either compromised peak efficiency [7] or limited efficiency improvement at PBO [9], largely due to the lossy mm-Wave Doherty power combiners and imperfect Doherty operations. Outphasing is an alternative and popular solution to achieve large PBO efficiency enhancement while maintaining a high peak efficiency. A typical outphasing architecture is composed of two PAs that operate with saturated envelope but dynamically varying phase difference, to synthesize the desired envelope varying complex modulation signals (Fig. 1). Due to the saturated PA operations, outphasing TXs maximize the output power (P_{out}) range toward the saturated output power (P_{sat}) and thus maximize the TX efficiency [13]–[22]. Moreover, rapid advance in baseband processors using deeply scaled CMOS reduces digital computation power for outphasing signal generations. In large-scale array systems, one modulator can also be shared among multiple TX/PA units to further reduce the computation power. As a result, outphasing is an attractive and potential

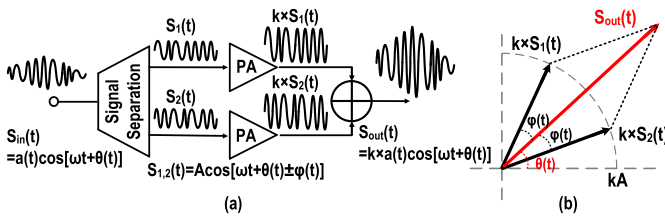


Fig. 1. (a) Outphasing architecture and (b) its vector representation.

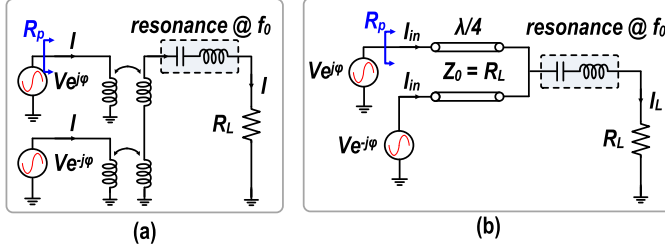


Fig. 2. Non-isolating outphasing combiner using (a) transformers and (b) $\lambda/4$ transmission lines.

candidate for 5G TX/PA architectures to achieve large-scale, energy-efficient, and agile beamforming.

However, a major remaining challenge is the lack of non-isolating outphasing power combiners at mm-Wave that have low loss and support desired active load modulation behavior, making high performance mm-Wave outphasing TX/PA still a challenging task [17], [22]. In essence, non-isolating outphasing combiners are series power combining networks. Two general outphasing combiner schematics are shown in Fig. 2. The series power combining nature of outphasing performs voltage summation at the output, which are typically implemented as transformers or $\lambda/4$ transmission line-based combiners (Fig. 2). Transformers at mm-Wave introduce significant leakage inductance and parasitic capacitance, making their behaviors as well as impedance balancing far from ideal series power combiners. In addition, limited quality factor makes the transformer-based outphasing combiner quite lossy at mm-Wave (e.g., 1.2 dB in [17]). As a result, the desired outphasing active load modulation is largely compromised, which leads to marginal efficiency improvement at PBO. On the other hand, $\lambda/4$ transmission line-based series power combiners occupy a significant chip footprint at low/medium mm-Wave (28 GHz), which is not practical for an on-chip implementation. Therefore, a new type of high-efficiency series power combiner that support desired active load modulation is in great need to realize high-performance mm-Wave outphasing TX/PA.

To address this difficulty, a 28-GHz packaged Chireix TX solution is proposed with direct on-antenna outphasing load modulation [23]. The Chireix TX contains two 28-GHz phase-shifted sub-TXs simultaneously driving a dual-feed loop antenna (DLA) on package (Fig. 3), which achieves high-efficiency on-antenna outphasing load modulation. This paper will show that a DLA naturally serves as a non-isolating outphasing power combiner, demonstrating superior active load modulation and low loss over conventional on-chip outphas-

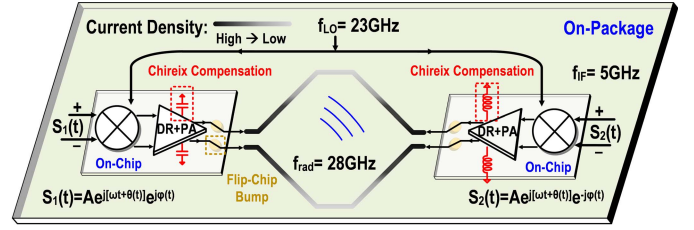


Fig. 3. Top-level schematic of the Chireix TX.

ing passive networks. Moreover, DLA offers built-in antenna impedance down-scaling at each feed due to its series power combining nature and greatly simplifies the matching interface between antenna and PAs [24]–[26]. To further improve the TX efficiency and its dynamic range, Chireix compensation and a hybrid outphasing operation are also incorporated into this design [27], [28], whose details will be presented in this paper.

II. DESIGN DETAILS OF THE 28-GHZ CHIREIX TRANSMITTER

The proposed two-way Chireix TX contains two 28-GHz phase-shifted sub-TXs driving one DLA on package. Each Chireix sub-TX circuit consists of an up-conversion double-balanced mixer and a two-stage PA (Fig. 3). The two sub-TX chips are integrated with the DLA by flip-chip packaging for minimized interconnection parasitic [29], [30]. With antenna serving as both a radiator and a power combiner, direct on-antenna power combining, and unique antenna-level active load modulation are realized. The corresponding design methodology and detailed circuit implementation are presented in this section.

A. Ideal Non-Isolating Power Combiners

Non-isolating power combiners introduce interactions between the two outphasing paths to achieve active load modulations for efficiency enhancement at PBO [13], which cannot be realized by isolating power combiner. As two popular examples, the transformer-based and $\lambda/4$ line-based non-isolating combiners are shown in Fig. 2. In Fig. 2(a), when driven by two outphased voltage sources ($Ve^{j\varphi}$ and $Ve^{-j\varphi}$), the output currents of the power devices at f_0 can be obtained as

$$I = \frac{Ve^{j\varphi} + Ve^{-j\varphi}}{R_L} = \frac{2V\cos(\varphi)}{R_L}. \quad (1)$$

The output power (P_L) and parallel load resistance (R_P) for each PA branch are derived as

$$P_L = \frac{1}{2}I^2R_L = \frac{2V^2\cos^2(\varphi)}{R_L} \quad (2)$$

$$R_P = \frac{1}{\text{Re}\left(\frac{I}{Ve^{\pm j\varphi}}\right)} = \frac{R_L}{2\cos^2(\varphi)}. \quad (3)$$

Thus, load resistance R_P increases at PBO when φ increases from 0° to 90° , which describes the desired outphasing active

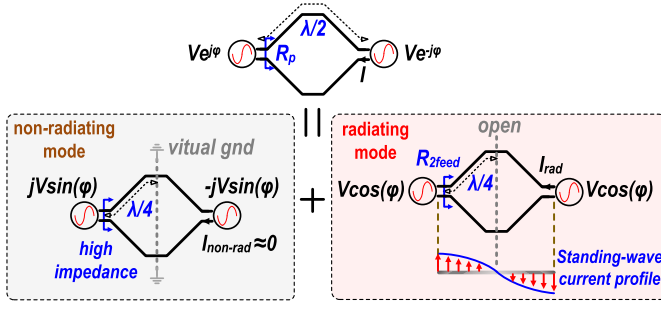


Fig. 4. Proposed on-antenna outphasing power combiner.

load modulation and thus ensures PBO efficiency enhancement.

The same conclusion can be drawn using the $\lambda/4$ combiner network in Fig. 2(b). The $\lambda/4$ T-lines convert the input voltage $Ve^{\pm j\varphi}$ into current $-jVe^{\pm j\varphi}/Z_0$. The currents from the two branches are then summed at the load. Alternatively, one can recognize that the two $\lambda/4$ T-lines and a shunt branch is equivalently a series power combiner. The resulting load current through R_L is

$$I_L = -\frac{jVe^{j\varphi} + jVe^{-j\varphi}}{Z_0} = -j\frac{2V\cos(\varphi)}{R_L}. \quad (4)$$

Therefore, the load voltage is expressed as

$$V_L = -j2V\cos(\varphi). \quad (5)$$

The currents injected into the combiner network can be determined as

$$I_{in} = j\frac{V_L}{Z_0} = \frac{2V\cos(\varphi)}{R_L} \quad (6)$$

which follows the same relationship in (1). As a result, the active load modulations described in (2) and (3) are realized in this case as well.

B. On-Antenna Outphasing Active Load Modulation

Despite the desired operation of idealistic non-isolating outphasing power combiners (Fig. 2), the absence of a low-loss effective mm-Wave non-isolating power combiner in practice largely impedes the implementation of outphasing architectures. To address this challenge, a DLA is exploited as an outphasing power combining network. When driven by two outphasing signals, common- and differential-mode analyses can be applied to study the proposed network (Fig. 4).

The differential-mode components ($\pm jV\sin\varphi$) of the input signals ($Ve^{\pm j\varphi}$) result in the non-radiating mode of the antenna. A virtual ground is formed in the middle of the DLA due to the differential excitations. The $\lambda/4$ arms of the antenna then transform the virtual ground to the high impedance (3.4 k Ω in simulation) at the antenna feeds. As a result, negligible RF current flows into the network in this undesired non-radiating/differential mode.

On the other hand, the common-mode signals ($V\cos\varphi$) of the outphasing sources excite the desired radiating mode of the antenna, and lead to low input impedance R_{2feed} . The total RF

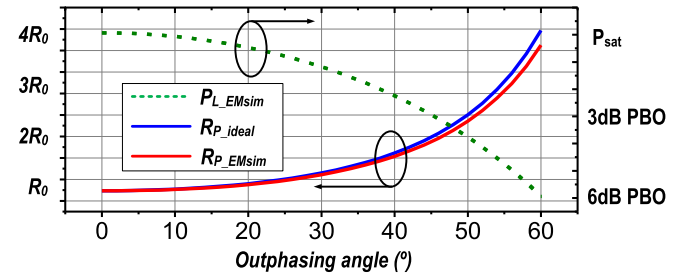


Fig. 5. Comparison of active load modulation between EM-simulated on-antenna outphasing network and ideal power combiner.

current on the antenna is thus only determined by the desired radiating common mode, and is described as

$$I = I_{rad} + I_{non-rad} \approx I_{rad} = \frac{V\cos(\varphi)}{R_{2feed}}. \quad (7)$$

In addition, DLA offers built-in antenna impedance down-scaling at each feed due to its series power combining nature ($R_{2feed} = R_{rad}/2$) [24]. Hence (7) can be further expressed as

$$I = \frac{2V\cos(\varphi)}{R_{rad}} \quad (8)$$

where R_{rad} is the radiation impedance of an identical loop antenna but with a standard single feed. As a result, the P_L and R_P in the proposed DLA can be obtained as

$$P_L = 2 \times \frac{1}{2} I^2 R_{2feed} = \frac{2V^2\cos^2(\varphi)}{R_{rad}} \quad (9)$$

$$R_P = \frac{1}{\text{Re}\left(\frac{I}{Ve^{\pm j\varphi}}\right)} = \frac{R_{rad}}{2\cos^2(\varphi)}. \quad (10)$$

Comparing (9) and (10) with (2) and (3), the DLA achieves the same load modulation behavior as an ideal non-isolating outphasing power combiner, with the load R_L in the ideal conventional combiner replaced by the radiation impedance R_{rad} in the proposed DLA power combining structure.

This theoretical derivation is further verified by 3-D EM simulation results in Fig. 5. The active load modulation achieved using the proposed DLA as the combiner (R_{P_EMsim}) matches well with that realized by the ideal power combining network (R_{P_ideal}). The small discrepancy between R_{P_ideal} and R_{P_EMsim} is due to the finite differential-mode driving impedance of the DLA. Thus, the normalized power delivered to the antenna (P_{L_EMsim}) closely follows the theoretical $\cos^2\varphi$ relationship as indicated in (2) and (9), shown in Fig. 5.

The antenna EM structure, simulated antenna pattern, driving impedance at each feed (Z_{in}), and bandwidth are shown in Fig. 6. The DLA is designed on a 20-mil-thick RO4350 substrate with dimensions of 3.5 mm by 2.5 mm [Fig. 6(a)]. Including the flip-chip transition, the on-package DLA is designed to radiate at the target 28 GHz, and offer radiation impedance as the PA optimum load impedance R_{opt} , which directly simplifies the PA output matching.

The impact of packaging variations on the antenna impedance is also simulated and plotted in Fig. 7(a). The nominal dimension of a flip-chip bump is 50 μm in both diameter and height. The desired common-mode impedance

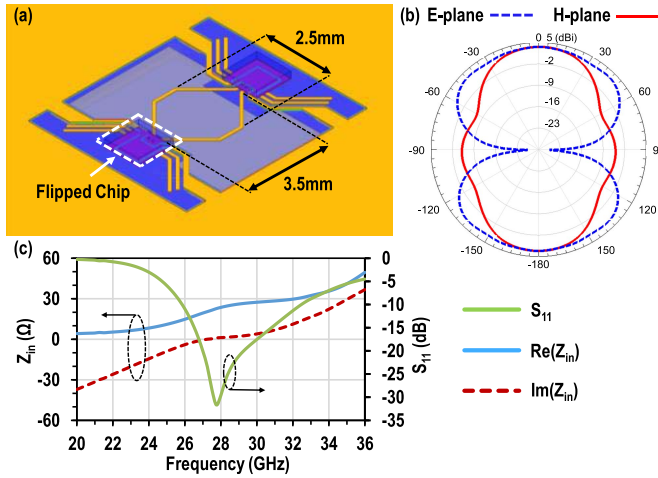


Fig. 6. (a) Dual-feed antenna EM model. (b) Antenna radiation pattern at $\phi = 0^\circ$. (c) Antenna input impedance and impedance bandwidth.

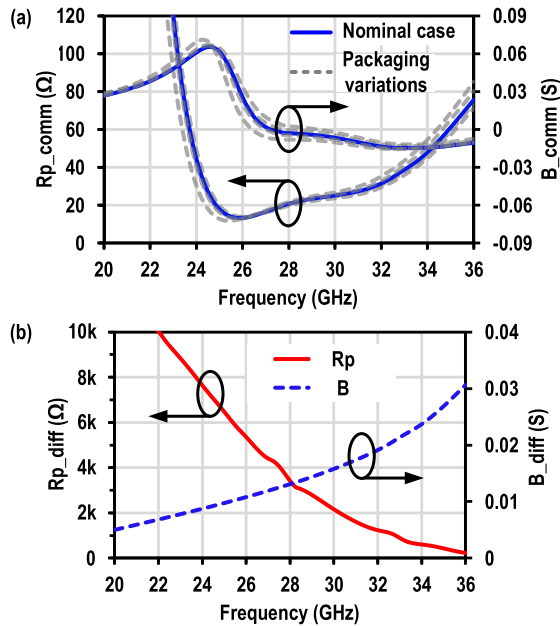


Fig. 7. (a) Desired common-mode antenna impedance under packaging variations. (b) Differential-mode antenna impedance.

remains relatively constant, considering $50\text{-}\mu\text{m}$ variation in the bump diameter ($50\text{--}100\text{ }\mu\text{m}$) and height ($30\text{--}80\text{ }\mu\text{m}$).

The antenna radiation/passive efficiency is 95% at 28 GHz, with a boresight antenna gain of 3.1 dBi [Fig. 6(b)]. The gain value is later used to calibrate the PA output power. The EM simulated antenna common-mode single-ended impedance is $23\text{ }\Omega$ at 28 GHz [Fig. 7(a)], while the differential-mode impedance is $3.4\text{ k}\Omega$ [Fig. 7(b)], verifying that the differential-mode is the non-radiation mode.

C. Antenna Response to Outphasing

Since Sections II-A and II-B show that the on-antenna power combiner achieves identical outphasing active load modulation as an ideal non-isolating power combiner,

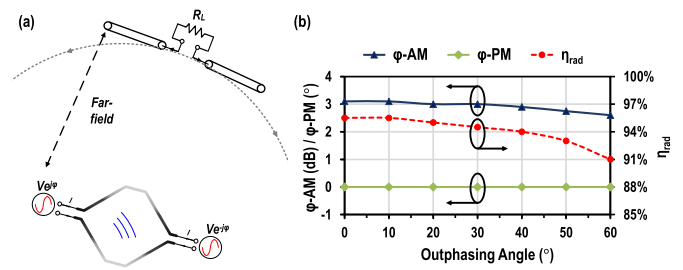


Fig. 8. (a) Far-field nonlinearity characterization setup of the on-antenna combiner. (b) Simulated amplitude and phase response to outphasing operation.

the antenna far-field behaviors under the outphasing driving conditions becomes the next question to answer. Because any change in the antenna characteristics (such as pattern and gain) induced by the outphasing operation will corrupt the demodulated signals at the far-field receiver (RX) end.

Unlike conventional TXs whose output power is combined and delivered to a fixed load, the multi-feed antenna processes the distributed transmitted power from its feeds and finally radiates it out. Since the on-antenna processing is not performed independently at each antenna feed, circuit performance parameters (P_{out} , AM-AM, AM-PM, etc.) should be examined at the target RX antenna load in the far-field, so that the complete on-antenna operations and channel non-idealities are captured. Since the input AM is decoded as the offset angle ϕ in outphasing, a simulation test setup [Fig. 8(a)] is built to quantify the ϕ -AM and ϕ -PM linearities of the outphasing TX chain. A dipole antenna is implemented as the target far-field RX, with the DLA driven by the outphasing signals as the TX.

Ideally, the antenna pattern and gain should remain unaffected during the outphasing operation to maintain large-signal linearity at the far-field RX. The simulated ϕ -AM and ϕ -PM results are shown in Fig. 8(b). To accurately quantify the nonlinearities only caused by the proposed DLA, outphased voltage sources are used in this simulation to drive the DLA. Within 60° outphasing angle, the ϕ -AM variation is below 0.5 dB, with ϕ -PM remaining constant. The degraded ϕ -AM variation beyond 60° outphasing angle is mainly caused by the unwanted differential-mode radiation. The antenna radiation efficiency (η_{rad}) is also plotted in Fig. 8(b), showing $>90\%$ η_{rad} during outphasing operation. It is worth noting that the large-signal linearity is preserved over the entire antenna field of view (FOV). Unlike the spatial IQ [31] and spatial outphasing technique [32], the proposed on-antenna outphasing maintains spatially independent signal integrity, which is further demonstrated in the measurement (Section III-B).

In this design, we introduce a hybrid outphasing scheme (Section II-F) to keep the outphasing angle below 60° and minimize ϕ -AM distortion induced by the antenna. Linear amplification is employed at deep PBO to improve the TX dynamic range and avoid the deep outphasing region ($60^\circ < \phi < 90^\circ$), since perfect matching between the branches and thereby complete out-of-phase component cancelation is difficult to achieve in practice [33], which often constrains the dynamic range in conventional outphasing TXs.

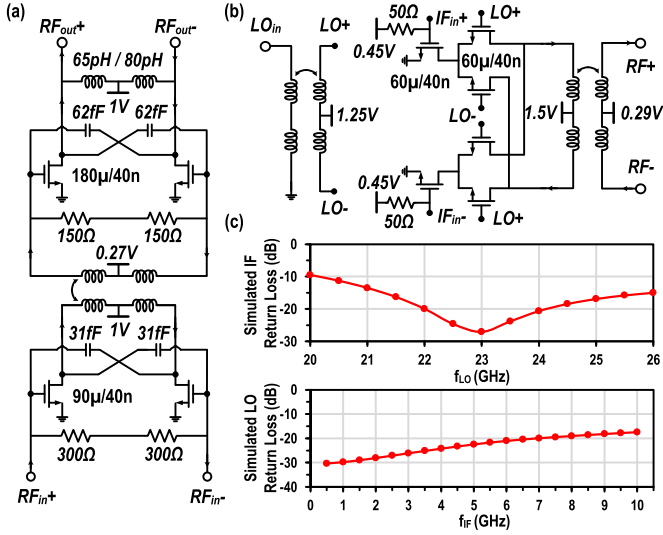


Fig. 9. (a) Schematic of the two-stage TX PA. (b) Schematic of TX mixer. (c) Simulated mixer LO and IF return loss.

D. TX Circuit Schematic

Each Chireix sub-TX circuit consists of an up-conversion double-balanced mixer and a two-stage PA. The antenna-based outphasing network combines the two paths and realizes active load modulation. The PA schematic is shown in Fig. 9(a). Each PA path comprises a driver stage and a PA stage with capacitive neutralization to boost power gain and enhance device stability [34], [35]. High-Q differential inductors are added at the PA output to resonate out device capacitance and provide supply bias and Chireix compensation. The details of Chireix compensation is discussed in Section II-E.

Double-balanced mixers up-converts the 5-GHz IF signals to the 28-GHz carrier. On-chip mixer implementation minimizes the phase/amplitude mismatch between the two outphasing paths. Fig. 9(b) shows the mixer schematic with transformers as the inter-stage matching and LO balun. The simulated wideband matching is plotted in Fig. 9(c). With 3-dBm LO driving strength, the mixer requires -7.5 -dBm IF input power to fully saturate the PA.

Although typical outphasing PAs/TXs are analyzed assuming voltage sources driving (Fig. 2), typical linear mm-Wave PAs exhibit appreciable output impedance at the operation frequency, leading to extra distortion and peak/PBO efficiency decrease in outphasing operations. An overdriven Class-B common-source (CS) PA is chosen as the PA stage for its smaller output impedance and inherently higher device efficiency. Operating power gain (G_p) and large-signal output impedance (R_{out}) of a CS and Cascode PA are compared in Fig. 10(a). In both cases, the same device size is used, and the two PAs are loaded with their corresponding load-pull impedances.

At 3-dB compression point (where the device is largely saturated), the output impedance of the CS PA is $10\ \Omega$ that is much smaller than that of the cascode PA ($39\ \Omega$). Due to the low device output impedance of an overdriven CS PA, and its constant output voltage amplitude constrained by the transistor saturation, it is a good approximation of a voltage-source, desired for PBO efficiency enhancement in

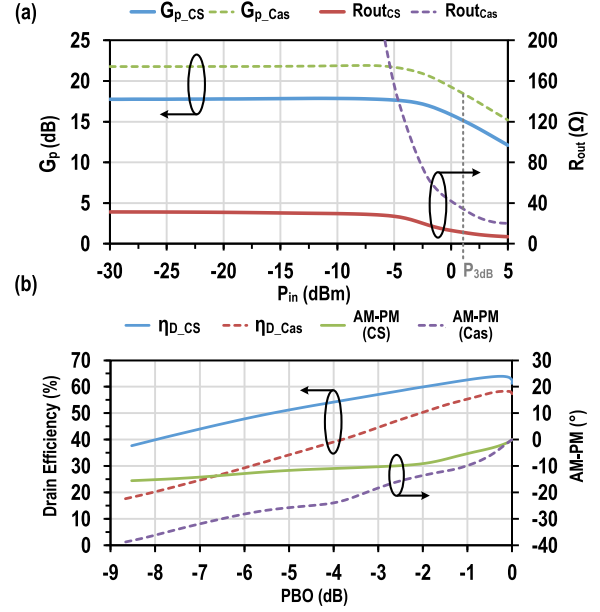


Fig. 10. (a) G_p and R_{out} of CS and Cascode PA versus input power. (b) Performance comparison between CS and cascode PA in outphasing operation.

outphasing architecture. The performance comparison between the CS and cascode PA in outphasing operations is summarized in Fig. 10(b). The PAs are driven at the same constant input power level ($P_{in} = 2$ dBm) with increasing outphasing angle for PBO in the two cases. The implementation with the CS PA not only exhibits a higher peak efficiency but maintains a higher efficiency at PBO. Moreover, much less AM-PM distortion is observed with the CS PA, further demonstrating the advantage of using overdriven CS topology in outphasing. In this design, the PA is driven at the 3-dB compression point for maximum efficiency during the outphasing operation. The driver stage is biased in Class-AB and has half size of the PA stage to provide sufficient driving strength to saturate the PA.

E. Chireix Compensation

During the outphasing operation, not only the real load resistance R_p is being modulated, but also an associated reactive component is varying versus the outphasing angle [Fig. 11(a)]. Revisiting the outphasing schematic in Fig. 2, the reactive parts seen by each branch can be written as

$$\text{Im}(Y) = \text{Im}\left(\frac{I}{V e^{\pm j\phi}}\right) = \pm \frac{\sin 2\phi}{R_L} \quad (11)$$

$$L_p = -\frac{1}{\omega \times \text{Im}(Y)} = \frac{R_L}{\omega \times \sin 2\phi}, \quad \text{for } \text{Im}(Y) < 0 \quad (12)$$

$$C_p = \frac{\text{Im}(Y)}{\omega} = \frac{\sin 2\phi}{\omega \times R_L}, \quad \text{for } \text{Im}(Y) > 0. \quad (13)$$

Therefore, one outphasing path sees an inductive load, while the other observes a capacitive load. It should be noted that the use of saturated linear PAs (such as Class-B) cannot cope with such power-dependent reactive load impedances, leading to substantial PBO efficiency degradation. Thus, Chireix compensations are needed to cancel out the reactive loads at a predefined outphasing angle for better efficiency enhancement

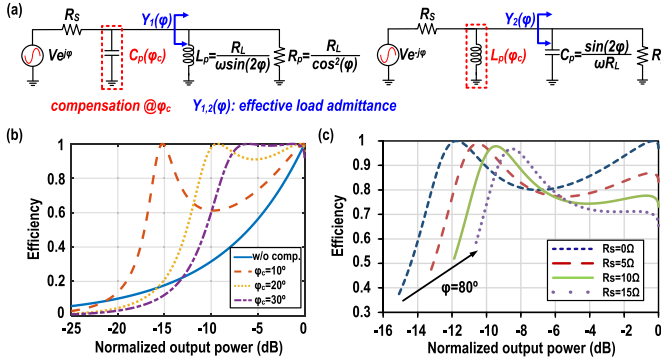


Fig. 11. (a) Equivalent circuit for the two outphasing paths. (b) Chireix combiner efficiency at power back-off with zero source resistance (R_S). (c) Chireix combiner efficiency at power back-off with different R_S values.

at PBO [14]–[16]. Theoretical PA efficiency (η) with a Chireix power combiner is derived as [14], [16], [20]

$$\eta = \frac{2\cos^2\varphi}{\sqrt{(2\cos^2\varphi)^2 + (\sin 2\varphi - \sin 2\varphi_c)^2}} \quad (14)$$

where φ_c is the predefined compensation angle. By substituting (2) into (14), the relationship between η and P_L is obtained and plotted in Fig. 11(b). Without any compensation, the PA efficiency rolls off rapidly during the PBO. By choosing different compensation angles (φ_c), the efficiency can be optimized at the corresponding PBO level.

The above analysis assumes ideal voltage sources with zero source resistance (R_S). For a practical design, the effect of R_S should be evaluated. Fig. 11(c) shows the influence of R_S on the efficiency for the same Chireix compensation angle ($\varphi_c = 15^\circ$) with lossless passives. All the four curves are plotted up to the PBO level with a maximum outphasing angle of $\varphi = 80^\circ$. Shown in Fig. 11(c), larger R_S value directly degrades the peak efficiency. Moreover, the second efficiency peak moves to a smaller PBO level with a larger R_S . In addition, the dynamic range is also reduced by a larger R_S for the same maximum outphasing angle. These observations further explain why a lower PA output impedance is preferred in an outphasing architecture, justifying the use of overdriven Class-B CS PAs. In this design, a 15° compensation angle is chosen to optimize the TX average efficiency when transmitting 64-QAM signals.

It is worth noting that the ideal symmetry of outphasing architecture no longer holds by adding the reactive Chireix compensation [15], [36]. This is due to the non-zero R_S of the voltage sources. In this case, the peak output power (0-dB PBO) is achieved at a shifted outphasing angle φ_0 , rather than at $\varphi = 0^\circ$. The output power is now given by a modified relationship as

$$P_L = \frac{2V^2}{R_L} \times \frac{1 + 2(R_S B_C)^2}{\left(1 + \frac{2R_S}{R_L} + R_S^2 B_C^2\right)^2} \cos^2(\varphi - \varphi_0) \quad (15)$$

where $\varphi_0 = \tan^{-1}(B_C R_S)$ and $B_C = \sin 2\varphi_c / R_L$. Compared with (2) for an ideal outphasing case, the relationship shown in (15) implies that constant phase shift and magnitude scaling should be considered for outphasing signal generation.

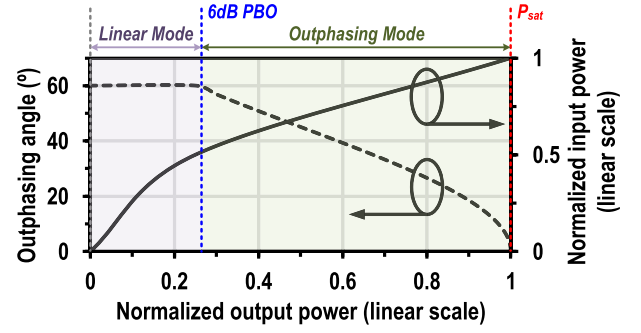


Fig. 12. Hybrid outphasing operation principle.

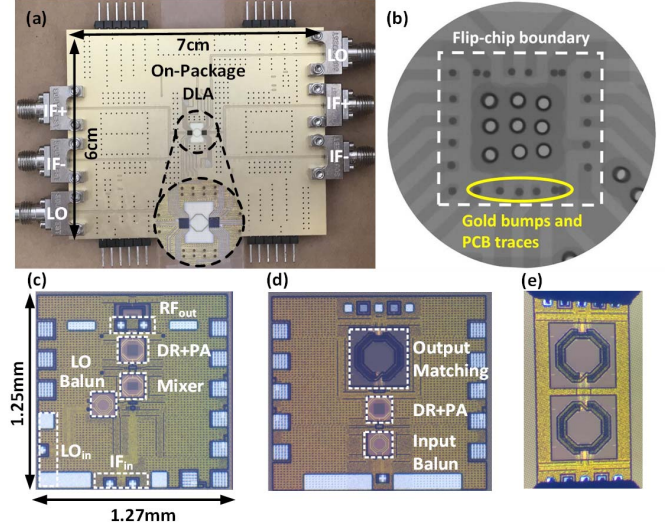


Fig. 13. (a) Fully packaged Chireix TX module. (b) X-ray image for verification of flip-chip transition alignment. (c) Sub-TX die photograph. (d) PA standalone TS. (e) Back-to-back TS of OMN of PA TS.

F. Hybrid Outphasing Operation

As presented in Section II-C, the outphasing angle is kept within 60° (corresponding to 6-dB PBO) for better linearity and dynamic range. Within this 6-dB PBO, the TX input power is reduced gradually. First, due to the increasing load at the increasing outphasing angle, less input power is actually required to drive the device into saturation [27], [37], [38]. Moreover, this gradually decreasing input power reduces the PA gain compression and improves the PA power-added efficiency (PAE). Beyond 6-dB PBO, the TX is operating in its linear operation mode where only input amplitude is being decreased with the outphasing angle fixed at 60° . The complete hybrid outphasing operation is illustrated in Fig. 12. Similar principles but with mixed-mode operations have also been applied in [27] and [28] for GHz implementations.

III. MEASUREMENT RESULTS

The 28-GHz Chireix TX IC chips are fabricated in GlobalFoundries 45-nm CMOS SOI process. Two sub-TX chips are flip-chip packaged with the on-package antenna. A separate PA test structure (TS) is also implemented in the same process for accurate power characterization. Fig. 13 shows the photograph of the fully packaged TX module, the X-ray image of the flip-chip transitions, and micrographs of the TX chip and PA TS.

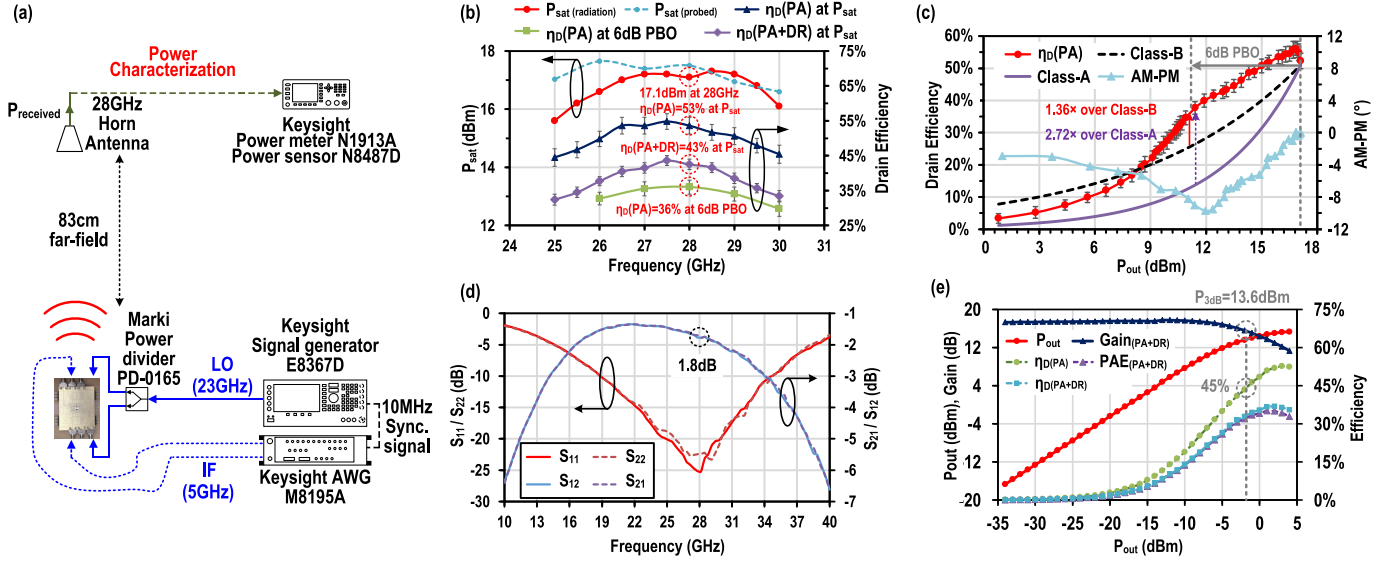


Fig. 14. (a) CW measurement setup. (b) Chireix TX performance over frequencies. (c) PA large signal behaviors at 28 GHz. (d) S-parameters of the back-to-back PA TS OMN. (e) PA TS large signal performance at 28 GHz.

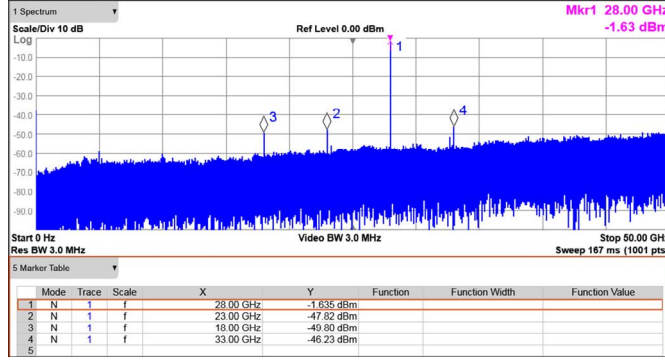


Fig. 15. Measured TX output spectrum.

A. Continuous-Wave Measurement

Fig. 14(a) depicts the CW measurement setup. Two channels from the arbitrary waveform generator (AWG) are used to generate the desired outphasing signals at IF (5 GHz), with a signal generator providing the LO (23 GHz) input for the two chips. A horn antenna is placed at the far-field (83 cm) to receive the radiated CW signal, and a power sensor measuring the received power (P_r). The Equivalent Isotropically Radiated Power (EIRP) of the TX can then be calculated by de-embedding the PL and RX antenna gain (G_{horn}), as

$$\text{EIRP} = P_r - G_{\text{horn}} + \text{PL}. \quad (16)$$

The output power (P_{out}) of the Chireix TX is defined as the total power delivered to the antenna. Two approaches are employed to achieve accurate Chireix TX P_{out} measurement.

It can be attained by subtracting the TX outphasing antenna gain (G_{ant}) from the EIRP, as

$$P_{\text{out}} = \text{EIRP} - G_{\text{ant}}. \quad (17)$$

The P_{out} based on the measured EIRP and EM simulated G_{ant} is plotted in Fig. 14(b), yielding a 17.1-dBm P_{sat} at 28 GHz

with 53% PA-stage drain efficiency (DE) (η_D). Including the driver power consumption, the total PA $\eta_D(\text{PA}+\text{DR})$ is 43% ($\approx \text{PAE}$, considering $>15\text{-dB}$ two-stage PA power gain). The P_{sat} 1-dB bandwidth covers from 25.5 to 30 GHz, while maintaining at least 45% $\eta_D(\text{PA})$ and 34% $\eta_D(\text{PA}+\text{DR})$. The PA large-signal behaviors at 28 GHz are summarized in Fig. 14(c). Four independent measurements on two different TX modules have been performed. Based on these measurements, the standard deviation on the measured PA DE is then derived, which is reflected as the error bars in both Fig. 14(b) and (c). At 6-dB PBO, 36% $\eta_D(\text{PA})$ is achieved, demonstrating $1.36\times/2.72\times$ efficiency enhancement over an idealistic class-B/class-A PA.

Alternatively, the TX P_{out} is obtained from the measurement of the PA TS with direct probing. The same PA core used in the TX is implemented as a separate TS with an additional output matching network (OMN) transforming differential 100 Ω to the PA optimum load impedance ($R_{\text{opt}}//L_{\text{opt}}$), where R_{opt} is also the antenna input impedance. As shown in Fig. 14(e), the PA TS achieves 13.6 dBm P_{sat} with 45.4% $\eta_D(\text{PA})$ at 28GHz. The P_{sat} is defined at the 3-dB gain compression point, since the two PAs are driven to the same saturation level during the outphasing operation. Since the PAs directly interface with the antenna in the TX design, realizing on-antenna power combining, the loss from the PA TS OMN should be de-embedded for P_{out} calculation. A back-to-back OMN TS is also implemented [Fig. 13(e)] to capture its loss, and the measured S-parameters of the OMN are shown in Fig. 14(d). At 28 GHz, the passive loss of the back-to-back OMN TS is 1.8 dB, indicating that the passive loss of a single OMN TS is 0.9 dB. After de-embedding the loss from the OMN, the PA achieves 14.5-dBm P_{sat} , with 55.8% $\eta_D(\text{PA})$. Therefore, considering 3-dB output power increase due to the two-way combining, the P_{sat} based on TS measurement is 17.5 dBm, matching well with the radiation-based P_{sat} value of 17.1 dBm at 28 GHz. The comparison in measured P_{sat} between the radiation-based and probing-based method is also

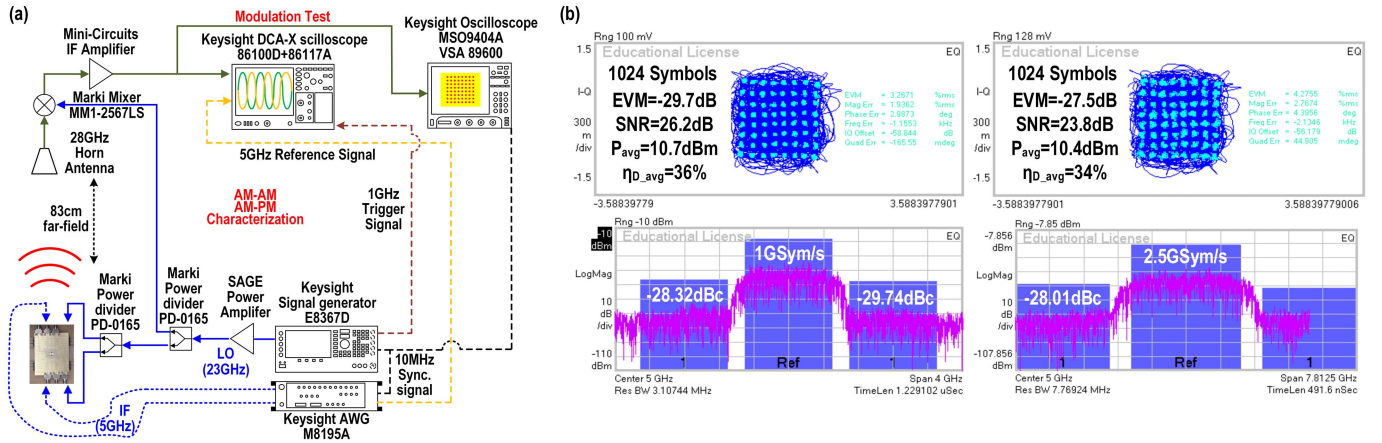


Fig. 16. (a) Nonlinearity characterization and modulation test setup. (b) Measurement results of 64-QAM at 6 and 15 Gb/s.

shown in Fig. 14(b). Overall, the two methods achieve a very close agreement on the measured P_{sat} around 28 GHz. In addition, the radiation-based P_{sat} shows a smaller value at lower frequency range, due to the limited antenna bandwidth [Fig. 6(c)], compared to a wideband transformer-based OMN [Fig. 14(d)].

The TX output spectrum is measured and shown in Fig. 15. A horn antenna (SAR-2013-34-S2) and a wideband LNA (RLNA16G32G) are used to receive and amplify the signals before the spectrum analyzer. Except the desired carrier frequency at 28 GHz, the LO leakage (23 GHz), image signal (18 GHz), and IF 2nd harmonic up-conversion (33 GHz) are identified from 18 to 33 GHz. Both the horn antenna and the LNA have around 3-dB gain variation within the same frequency range, which is readily de-embedded for the measurement of the undesirable signal rejection. At least 40-dB rejection is achieved for the unwanted tones mentioned above, which prevents the corruption of the demodulated signals.

B. Wireless Modulation Test

To ensure high-performance modulations, the TX large-signal nonlinearities (AM-AM/AM-PM) are first characterized. The measurement setup for AM-AM and AM-PM is shown in Fig. 16(a). Three channels from AWG are used, with two of them providing the input two-path outphasing signals, and the third channel generating the reference signal for the phase comparison. A signal generator and an external power amplifier (PA) are employed to feed the LO power for both up- and down-conversion. The transmitted signal is picked up by the RX horn antenna and then down-converted to the IF (5 GHz) by the mixer. The IF signal is amplified, sent to the sampling oscilloscope, and compared with the reference IF signal from the AWG for amplitude/phase nonlinearity characterization.

By applying the hybrid input conditions (P_{in} , ϕ) shown in Fig. 12, the measured large-signal behaviors are summarized in Fig. 14(c). A memoryless 2-D (P_{in} , ϕ) lookup table (LUT) is used to generate the desired outphasing input signals and perform digital pre-distortion (DPD) to correct the AM-AM/AM-PM errors. Compared with existing outphasing PAs/TXs [17]–[22], no additional computation resource is required.

The wireless modulation measurement setup is shown in Fig. 16(a). Essentially, the same setup is used only with the sampling oscilloscope replaced by a mixed signal oscilloscope for digital demodulations. As shown in Fig. 16(b), the Chireix TX achieves -29.7 -dB EVM with 36% average PA DE (η_{D_avg}) for 6 Gb/s (1GSym/s) 64-QAM signal, and -27.5 -dB EVM with 34% η_{D_avg} for 15 Gb/s (2.5 GSym/s) 64-QAM signal, demonstrating state-of-the-art modulation bandwidth and average efficiency. The EVM is referenced to the peak amplitude of the constellation. The TX performance in a wireless testing environment with 64-QAM at different symbol rates is further summarized in Fig. 17.

PA/TX architectures with spatial power combining such as the Spatial IQ Combiner TX [31] and Spatial Outphasing TX [32] are demonstrated in earlier literature. However, these designs exhibit substantially narrower FoV than the intrinsic antenna FoV, due to their vectorial spatial signal combining. Since the desired outphasing or IQ combining only happens at a specific angle, the signals will be severely distorted when the TX and RX are mis-aligned [31], [32]. Instead, the proposed Chireix TX achieves on-antenna outphasing combining before its radiation and therefore ensures undistorted signals over the full antenna FoV. Fig. 18 depicts the modulation results of the Chireix TX over various radiation directions. When the receiving angle is swept from -60° to $+60^\circ$, the measured demodulated signals exhibit consistent and angle-independent performance with consistent EVM values over the entire antenna FoV. Such angle-independency is essential for establishing reliable wireless communication links in practical usage scenarios that prefer easy and robust TX/RX alignment. Therefore, our on-antenna outphasing TX is particularly suitable for constructing phased arrays or MIMO systems with wide array scanning angle, which in contrast cannot be achieved by reported Spatial IQ Combiner TX [31] or Spatial Outphasing TX [32] with inherent narrow element FoV.

C. PA Reliability Testing

To evaluate the reliability of the CS PA with 1-V supply biasing ($V_{\text{dd}} = 1$ V), RF stress testing was performed continuously over a cumulative 24-h period under CW operation. The CS PA TS is used in this test, so that its output power can be directly probed for measurement.

TABLE I
COMPARISON WITH STATE-OF-THE-ART MM-WAVE SILICON-BASED PAs/TXs

	This Work	[22] Rabet ISSCC '18	[17] Zhao JSSC '12	[39] Shakib ISSCC '17	[7] Hu ISSCC '17	[40] Park TMTT '16	[41] Ali ISSCC '18	[42] Li ISSCC '18
Technology	45nm CMOS SOI	130nm SiGe	40nm CMOS	40nm CMOS	130nm SiGe	28nm CMOS	65nm CMOS	130nm SiGe
Architecture	On-antenna outphasing	Outphasing with on-chip passive	Outphasing with on-chip passive	Analog	Doherty	Analog	Analog with AM-PM correction	Continuous-mode harmonically tuned
Supply (V)	1.0	4.0	1.0	1.1	1.5	2.2	1.1	1.9
Frequency (GHz)	28	28	60	27	28	28	28	28.5
P_{linear} (dBm)	17.1	23	15.6	13.7 ($P_{1\text{dB}}$)	15.2 ($P_{1\text{dB}}$)	18.6 ($P_{1\text{dB}}$)	14 ($P_{1\text{dB}}$)	15.2 ($P_{1\text{dB}}$)
η @ P_{linear}	PA: 53% (DE_{avg}) PA + DR: 43% (PAE_{avg}) $\pm 2.4\%$ (σ) $\pm 1.6\%$ (σ)	PA: 41.4% (PAE)	PA: 25% (PAE)	PA + DR: 31.1% (PAE)	PA: 28.3% (CE) PA + DR: 19.5% (PAE)	PA: 43.3% (PAE)	PA + DR: 34.7% (PAE)	PA: 43% (PAE) PA + DR: 39.2% (PAE)
η @ 6dB PBO	PA: 36% (DE_{avg}) PA + DR: 23% (PAE_{avg}) $\pm 2.1\%$ (σ) $\pm 1.5\%$ (σ)	PA: 34.7% (PAE)	PA: 9% (PAE) [†]	PA + DR: 12.5% (PAE) [†]	PA: 21% (CE) [†] PA + DR: 11% (PAE) [†]	PA: 20% (PAE) [†]	PA + DR: 11% (PAE) [†]	PA: 19% (PAE) [†] PA + DR: 16% (PAE) [†]
$2\eta_{\text{6dB}} / \eta_{\text{linear}}^*$	1.36 (DE)	1.67	0.72	0.8	1.48 (CE)	0.92	0.63	0.88
Modulation Type and Data Rate	64QAM, 6Gb/s 64QAM, 15Gb/s	64QAM OFDM, 0.48Gb/s	16QAM, 0.5Gb/s	64QAM OFDM with 8CC, 4.8Gb/s	64QAM, 6Gb/s	64QAM WLAN, 0.48Gb/s	64QAM, 2.04Gb/s	64QAM, 18Gb/s
EVM (dB)	-29.7 -27.5	-30.5**	-22	-25	-26.6	-27.5	-26.4**	-25
P_{avg} (dBm)	10.7 10.4	14.3	12	6.7	7.2	11.0	9.8	9.8
η_{avg}	PA: 36% PA + DR: 20.6%	PA: 25.3%	PA: 15%	PA + DR: 11%	PA: 14.4%	PA: 17.3%	PA + DR: 18.2%	PA + DR: 18.4%

[†] Graphically estimated.

[‡] Measurement standard deviation derived from four independent measurements on two different TX modules.

*Compared to an ideal class-B PA with the same efficiency at P_{linear} .

**Referenced to the RMS value of the constellation magnitudes ($EVM_{\text{RMS}} \approx \text{SNR}$) [44].

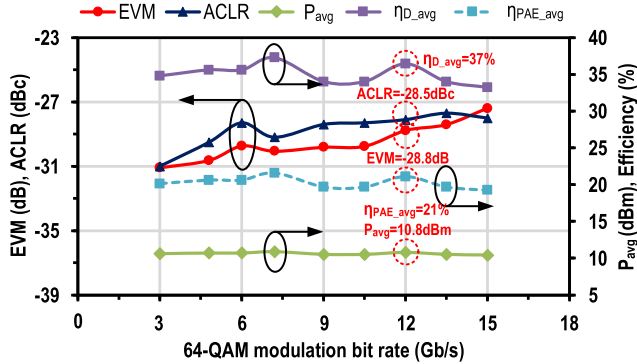


Fig. 17. Chireix TX modulation performance versus bit rate with 64-QAM.

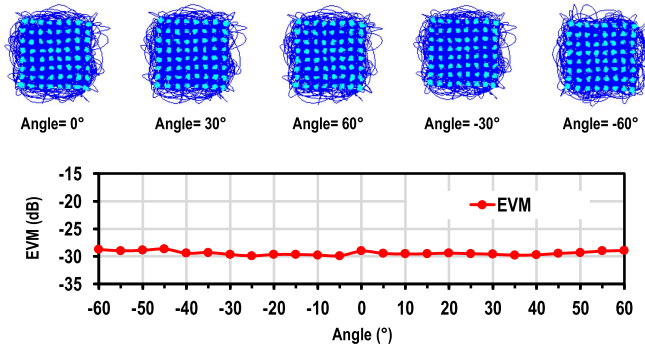


Fig. 18. Measured modulation results at different directions with 6 Gb/s 64-QAM, demonstrating wide antenna FoV.

The CS PA chip was driven at the input 1-dB compression point ($IP_{1\text{dB}} = -5$ dBm), with P_{out} and PA-stage dc current ($I_{\text{DC_PA}}$) being monitored. The reliability testing results are

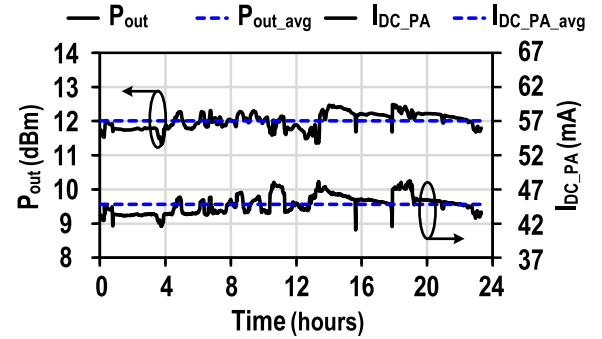


Fig. 19. PA reliability test at $P_{1\text{dB}}$ with P_{out} and $I_{\text{DC_PA}}$ being monitored over 24 h ($P_{\text{in}} = -5$ dBm, $V_{\text{dd}} = 1$ V).

summarized in Fig. 19, which reveals that no degradation on the PA performance is observed over continuous 24-h period. In practical applications especially for communication, the PA is mostly used to amplify complex modulated signals with certain PAPR. The average P_{out} during modulation is much less than $P_{1\text{dB}}$ (e.g., >6 dB for 64-QAM), which means the RF stress on the PA is largely relaxed in real applications compared to the performed CW reliability testing.

Table I shows the comparison of the proposed Chireix TX with the state of the art [43]. By exploiting a unique on-antenna outphasing architecture, the Chireix TX demonstrates high-efficiency and high-speed transmission of complex modulated signals in a wireless testing environment. Measurement proves that the proposed Chireix TX achieves high efficiency both at the peak output power and PBO. Modulation test further shows the TX manages to efficiently transmit up to 15 Gb/s 64-QAM signals in a wireless dynamic environment. In addition, the on-antenna active load modulation network can

be implemented with other baseline antennas, such as dipole, patch, spiral antennas. The DLA-based outphasing combiner demonstrated in this paper is only a proof-of-concept example.

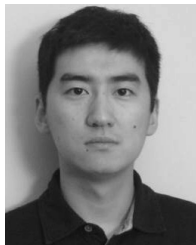
IV. CONCLUSION

This paper presents a Chireix TX architecture that explores multi-feed antennas to achieve unique on-antenna outphasing load modulation and demonstrate high-efficiency transmission of wideband modulated signals. High-efficiency power combining, outphasing load modulation and mm-Wave signal radiation have been realized in a single antenna structure, accomplishing multi-functional signal processing. More importantly, unlike other spatial combining techniques, the proposed Chireix TX preserves a wide element-level FoV. By leveraging DLA, a proof-of-concept Chireix TX demonstrates measurement performance that advances the state of the art.

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